predictions. This flexibility motivates and drives the design of GPyTorch, whose emphasis on modularity helps us extend and implement our own custom mean function effortlessly. We will see the same flexibility and modularity in GPyTorch's covariance functions, which we will discuss next.

3.4 Defining variability and smoothness with the covariance function

While the mean function defines our expectation of what the overall behavior of the objective function, the covariance function, or the kernel, of a GP plays a more complex role: defining the relationship between data points within the domain and controlling for the structure and smoothness of the GP.

In this section, we will compare how the predictions by a GP change as we vary different components of our model. From there, we will gain practical insights on how to select an appropriate covariance function for a GP model. The code that we will be using is in `CH03/02 - Covariance functions.ipynb`.

Throughout these examples, we will be using the Forrester function, which we saw in Section 2.4.1 of Chapter 2, as our objective. We once again randomly sample three data points between -3 and 3 and use them as our training data set. All predictions visualized in this section will be from a GP trained on these three points.

3.4.1 Setting the scales of the covariance function

The first way of controlling the behavior of a GP via its covariance function is to set the length scale and the output scale. These scales, just like the constant or the coefficients in the mean function, are the hyperparameters of the covariance function.

- A length scale controls the scale of the GP's input, and thus how fast the GP can change along an axis.
- The output scale defines the range of the GP's output, or in other words, the range of its predictions.

By setting these scales to different values, we can either increase or decrease the uncertainty in our GP's predictions, as well as scale the range of our predictions.

We will use the following implementation:

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